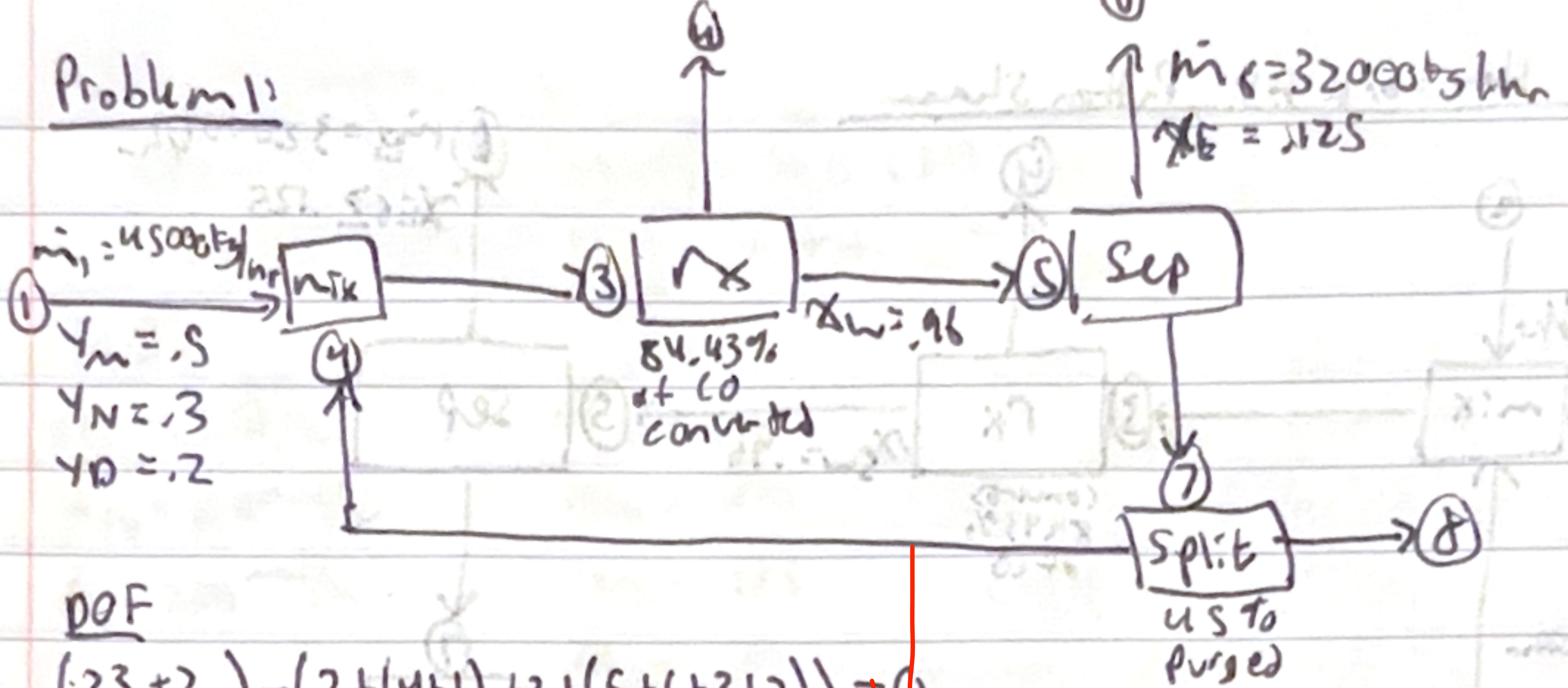
$$6CO + 3H_2O \rightarrow C_2H_5OH + 4CO_2 \quad (R1)$$
$$4CO + 2H_2O \rightarrow CH_3COOH + 2CO_2 \quad (R2)$$

Problem 1:



DOF

$(23+2) - (2+(4+1)+2+(5+6+3+2)) = 0$

Correctly specified.

$M_1 = (.50)(28) + (.20)(44) + .30(28) = 31.2$

$\dot{n}_1 = \frac{45000}{31.2} = 1442$ mol/hr

$\dot{n}_M = 721$ mol/h $\quad \dot{n}_{1D} = 289$ mol/h $\quad \dot{n}_{1N} = 433$ mol/hr

$\dot{m}_{6E} = 4000$ kg/h $\quad \dot{m}_{6W} = 28000$ kg/hr

$\dot{n}_{6E} = 86.96$ mol/h $\quad \dot{n}_{6W} = 1555.6$ mol/h

$\xi_1 = 86.96$ mol

CO: .8443 %

$\dot{n}_{5M} = (1-.8443) \cdot 721 = 112.3$ mol/hr

CO cons $= 721 - 112.3 = 608.9$

$6\xi_1 + 4\xi_2 = 608.9 \quad \xi_2 = 21.79$

CO_2: $4\xi_1 + 2\xi_2 = 391.4$ mol/h

$\dot{n}_{5D} = 289 + 391.4 = 679.9$ mol/h $\quad \dot{n}_{5N} = 433$ mol/hr

$$\frac{18\dot{n}_{5W}}{18\dot{n}_{5W} + 60\dot{n}_{5A}} = 0.96$$

$\dot{n}_{4A} = .55\dot{n}_{5A}$

$\dot{n}_{8A} = \dot{n}_{4A} + \xi_2 = .55\dot{n}_{5A} + 21.79$

$\dot{n}_{5A} = 48.42$ mol/hr

$\dot{n}_{5W} = 3873.6$ mol/hr

$\dot{n}_{8W} = (.45)(3873.6) = 1743$ mol/hr

$\dot{n}_{8A} = (.45)(48.42) = 21.79$ mol/hr

$\dot{n}_{4W} = (.55)(3873.6) = 2130.5$ mol/h

$\dot{n}_{4A} = (.55)(48.42) = 26.63$ mol/hr

$\dot{n}_{2W} = \dot{n}_{3W} - (3\xi_1 + 2\xi_2)$

$\dot{n}_{2W} = 3603$ mol/hr

Single pass

X_W reactor: $\frac{w \text{ cons}}{w \text{ react}} = \frac{3\xi_1 + 2\xi_2}{\dot{n}_{3W}} = \frac{304.45}{5733.6} = .053$

overall X_W overall $= \frac{304.45}{3603.1} = .0845$

overall conversion > single pass conversion

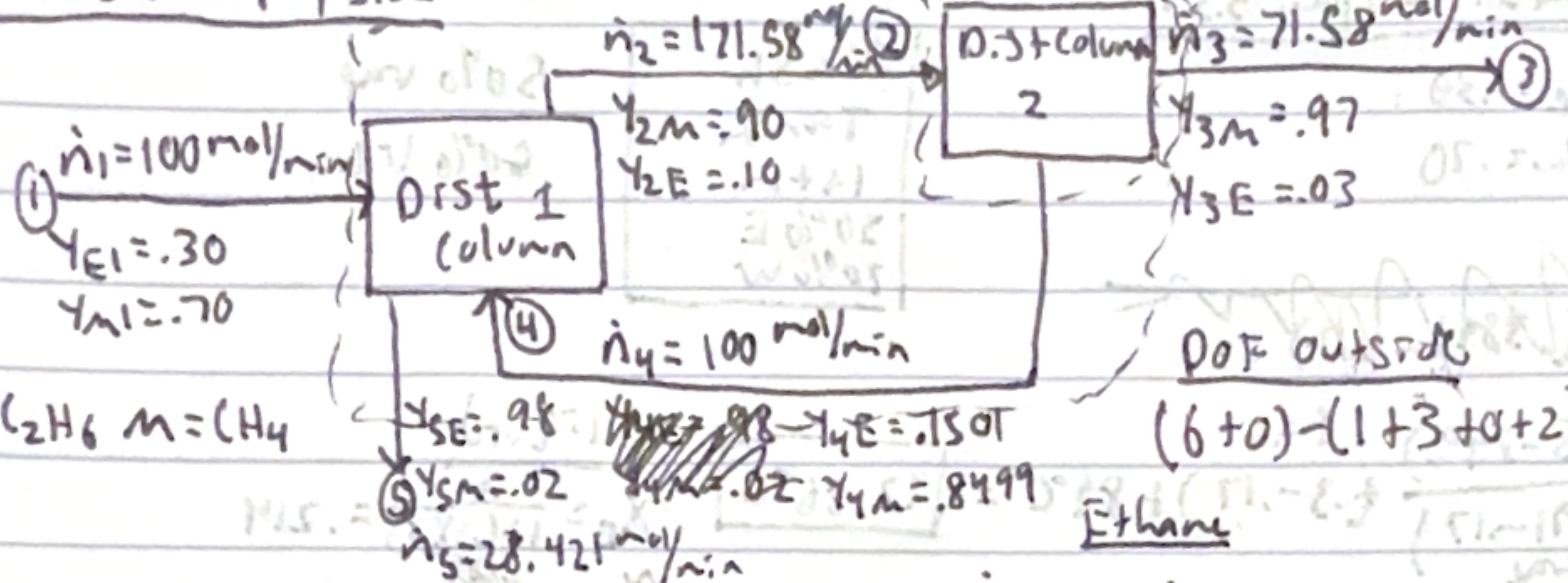
Improve:

You could improve the separation, reduce the purge fraction, recover water from the product stream & recycle it.

−4 pts

44/50

Problem 2: Murphy 5.32



$E = C_2H_6 \quad M = CH_4$

DOF outside

$(6+0)-(1+3+0+2)=0$ ✓

Dist 1

Ethane

$\underline{\dot{n}_4 y_{4E}} + \dot{n}_1 y_{E1} = \underline{\dot{n}_5} y_{5E} + \underline{\dot{n}_2} y_{2E}$

Methane

$\dot{n}_4 \underline{y_{4M}} + \dot{n}_1 y_{M1} = \underline{\dot{n}_5} y_{5M} + \underline{\dot{n}_2} y_{2M}$

~~answer~~

Dist 2 DOF: $(6+0)-(2+2+0+2)=0$

E: $\underline{\dot{n}_2}\, y_{2E} = \dot{n}_{3E} + \dot{n}_4 \underline{y_{4E}}$

M: $\underline{\dot{n}_2}\, y_{2M} = \dot{n}_{3M} + \dot{n}_4(1-\underline{y_{4E}})$

$1 - y_{4E} = y_{4M}$

$\dot{n}_2 = \frac{\dot{n}_{3E} + \dot{n}_4 y_{4E}}{y_{2E}}$

$\left(\frac{\dot{n}_{3E} + \dot{n}_4 y_{4E}}{y_{2E}}\right) y_{2M} = \dot{n}_{3M} + \dot{n}_4(1-y_{4E})$

$19.33 + 900 y_{4E} = 69.43 + 100 - 100 y_{4E}$

$y_{4E} = \frac{150.1}{1000} = .1501 \quad y_{4M} = .8499$

$\dot{n}_2 = \frac{2.15 + 15.01}{.10} = 171.58$ mol/min ✓

Ethane

$\dot{n}_1 y_{E1} = \underline{\dot{n}_5} y_{5E} + \underline{\dot{n}_3} y_{3E}$

M: $\dot{n}_1 y_{M1} = \underline{\dot{n}_5} y_{5M} + \underline{\dot{n}_3} y_{3M}$

$\frac{\dot{n}_{1E} - \dot{n}_3 y_{3E}}{y_{5E}} = \dot{n}_5$

$\dot{n}_{1M} = \left(\frac{\dot{n}_{1E} - \dot{n}_3 y_{3E}}{y_{5E}}\right) y_{5M} + \dot{n}_3 y_{3M}$

$70 = \left(\frac{30 - \dot{n}_3(.03)}{.98}\right).02 + .97\dot{n}_3$

$70 = (30.61 - \dot{n}_3 .0306).02 + .97\dot{n}_3$

$69.388 = .96939\dot{n}_3 \quad \dot{n}_3 = 71.58$ mol/min

$\dot{n}_5 = \frac{30 - 2.147}{.98} = 28.421$ mol/min

Fractional Recovery

$\frac{\text{mol recovered}}{\text{mol fed}}$

E: $\frac{(28.42)(.98)}{30} = 0.928$ ✗

M: $\frac{(71.58)(.97)}{70} = .992$ ✗ −4

26/30

Problem 3: Murphy 5.34

$y_e = .30$
$y_w = .70$

~~(.384)(-.17)~~

$$\frac{(95.5-89°C)}{(.3891-.17)}\text{ frac}\,(.3-.17)+89°C = \boxed{92.86°C}$$

flash Tank
1atm
30% E
70% W

50% vap
50% liq

liquid
$x_E = \frac{.3}{1+(.5)(.8)} = .214$
$x_w = .786$

$K_w = .78$
$K_E = 1.8$

Separation factor
$\frac{1.8\,E}{.78\,w} = \boxed{2.31}$

X -5

vapor
$x_E = (1.8)(.214) = .386$
$x_w = .614$

15/20